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PROJECT REPORT

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

TOPIC-CYBER CRIME DETECTION SYSTEMM

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INTRODUCTION

In the current digital environment, internet usage has grown exponentially, making communication faster and more accessible. At the same time, with these developments, cybercrime has increased manifold. People are receiving phishing messages, suspicious links, and scam calls, which try to fetch their sensitive information, like passwords, OTPs, and bank accounts, etc. At times, people, especially students and new internet users, are unaware of how to identify these risks, making them vulnerable to cybercrime attacks.

The objective of this project is to design a system using artificial intelligence, which helps people identify these risks and stay safe from cybercrime attacks. The system will incorporate basic concepts of machine learning, which will help identify suspicious messages, links, and phone numbers, etc. The system will be simple, interactive, and easy to use, while incorporating basic concepts of machine learning, which will help identify suspicious messages, links, and phone numbers, etc.

This project emphasizes the significance of digital awareness in today's digital environment.

PROBLEM STATEMENT:

With the rise of digital technologies, cyber crimes like phishing messages, harmful links, and scam phone calls have become extremely common. The average internet user, especially students, can’t identify these cyber crimes and becomes a victim of fraud because of their lack of knowledge.

Phishing messages, fake websites, and unknown phone numbers are used in cyber crimes. The existing security measures are complicated or require an internet connection, which is not easy for average internet users. Hence, there is a need to develop a simple yet efficient security tool.

KEY CHALLENGES INCLUDE:

- Difficulty in identifying phishing messages
- Lack of awareness about malicious links
- Risk of fraud from unknown phone numbers
- Limited access to simple cybersecurity tools

This project aims to solve these problems by developing a user-friendly system that detects threats and promotes safer digital practices.

OBJECTIVE:

The primary objective of this project is to spot cyber threats by checking digital messages for signs of phishing through basic artificial intelligence methods. Rather than just flagging obvious scams, it examines message wording carefully to catch hidden dangers. Checking web links comes next - each URL gets tested against known risky traits and common deceptive structures. Instead of relying solely on blacklists, the method uses pattern logic to judge if a site might be harmful. Phone number checks follow, scanning digits to uncover likely scam sources while guessing carrier and location details.

Beyond analysis features, attention goes toward how people interact with the system day-to-day. A step-by-step menu layout guides choices without requiring technical knowledge. Each function appears within reach, avoiding cluttered options or complex navigation paths. Through straightforward feedback, users learn about risks tied to specific inputs they provide. Focus stays on clarity, keeping results understandable even under uncertain conditions.

A primary aim involves showing ways fundamental ideas in artificial intelligence along with coding methods tackle everyday challenges. Built for clarity, the setup focuses on straightforwardness, performance, while remaining accessible - helping learners grasp core principles through hands-on exposure.

METHADOLOGY:

The project is implemented using a modular approach, where each component is responsible for detecting a specific type of cyber threat. Starting with how texts behave, harmful ones often repeat certain terms - those get marked stronger than ordinary phrases. Scoring happens by adding up these marks when a new message arrives, tipping the result toward risky or clear.

Instead of learning from data, one section watches web addresses closely, guided by fixed checks. Unusual lengths show up first, followed by outdated ways sites connect, too many separators, or names pretending to be banks. Each clue builds the decision without guessing. A check begins by confirming if the number follows correct formatting rules. Following that, guesses about carrier and location come from analyzing prefix data. Patterns such as repeating numbers or known fraud combinations raise red flags during review.

All components connect inside one interactive menu layout. Selection of a specific function delivers outcomes without delay.

FEATURES:

The cybercrime detection system offers several useful features that enhance its functionality and usability. From within the setup, a core piece stands out - an artificial intelligence tool trained to inspect messages, then judge if they're likely phishing attempts or harmless, using past examples.

Sitting alongside it, URL checks happen automatically, where links get scanned and ranked for danger through red flags in their structure. Instead of guessing, numbers go through a scanner too, revealing origin details like carrier and location while flagging potential threats. Built right into the flow, these tools sharpen accuracy without slowing things down. Starting off, there's a menu setup guiding people through various tools without confusion.

Moving along, outputs show risk levels alongside reasons, so it is obvious how each outcome came about. Then again, things like phone number transfers are factored in, which means results come across as approximations instead of fixed numbers. Because of this mix, the whole thing ends up being useful, clear, and straightforward to operate.

LIMITATIONS:

The system has some drawbacks that should be taken into account despite its benefits. The message detection model may not be as accurate as sophisticated machine learning models trained on large datasets because it is based on a small dataset and basic word scoring. The URL detection module's efficacy against sophisticated attacks may be limited because it relies on rule-based analysis and does not use real-time databases or external services to verify links.

Similar to this, the phone number analyzer only offers an estimated operator and region based on number prefixes, and because mobile numbers are portable, the results might not always be accurate. Additionally, the system doesn't have access to real-time threat intelligence, which could increase detection accuracy, and it runs offline.

Furthermore, compared to graphical applications, the text-based user interface might not be as aesthetically pleasing. However, considering the project's educational goal, these restrictions are acceptable.

FUTURE SCOPE:

The project can be further improved in several ways to enhance its functionality and accuracy. Integrating cutting-edge machine learning models, like natural language processing methods, to increase message detection accuracy is one potential improvement. For more efficient URL and phone number verification, the system can be linked to real-time APIs or online databases.

Creating a graphical user interface (GUI) with web technologies or tools like Tkinter would be another improvement, making the system more aesthetically pleasing and user-friendly. The AI model's performance can be enhanced by expanding the dataset used to train it to include a greater range of messages.

It is also possible to add extra features like real-time alert systems, password strength checks, and email analysis.These improvements would make the system more powerful and closer to real-world cybersecurity applications.

CONCLUSION

In conclusion, this project demonstrates how basic artificial intelligence and programming concepts can be applied to address real-world cybersecurity challenges.

The AI-based cybercrime detection system provides a simple yet effective solution for identifying phishing messages, malicious links, and suspicious phone numbers. By combining machine learning techniques with rule-based analysis, the system offers a practical approach to improving digital safety and awareness.

Although the system has certain limitations, it successfully highlights the importance of recognizing cyber threats and taking preventive measures. The project also emphasizes the value of learning and applying technical skills to solve real-life problems. Overall, this project serves as a useful educational tool and a foundation for developing more advanced cybersecurity solutions in the future.

REFRENCES

- Python Official Documentation – for understanding basic programming concepts and syntax
- Online tutorials and educational resources on phishing detection and cybersecurity awareness
- Articles on cyber fraud, phishing attacks, and online scams
- Information on mobile number portability and telecom systems
- General internet resources and study materials related to AI and machine learning basics